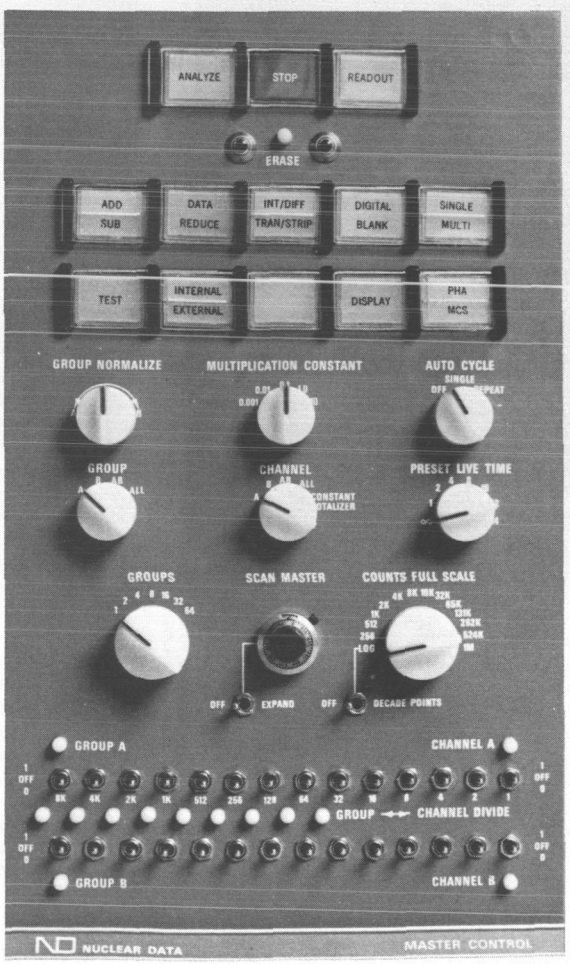
TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC.
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600



SERIES 3300 MASTER CONTROL

The Master Control Module contains facilities for establishing the primary operating modes of the Series 3300 System. In conjunction with the Data Handling Main Frame, it provides the facilities for all data processing functions. After data has been acquired, it can be subjected to operations such as spectrum stripping, integration, differentiation, multiplication, and transfer from one selected memory group to another.

Front panel controls enable digital selection of memory groups and channels for data storage, display, processing, and readout. Display intensification and digital blanking aid in determining area of interest location. The digital channel selection range is continuous down to one channel. Digital group size can be selected in binary steps down to 64 channels.

Logical programming for both pulse height analysis and multichannel scaling operation is another standard capability. In addition, a unique "Scan Master" feature enables stored data to be paraded through an expanded display window for closer examination. Only those data channels within the expanded window are addressed, providing a high scan rate regardless of window location.

Main Mode Selection

Illuminated pushbuttons establish the main operating modes of the system, i.e., Analyze, Stop, and Readout.

Sub-Mode Selection

Illuminated pushbuttons establish operational submodes subordinate to each main operating mode. These include:

DISPLAY — before, during, and after accumulation. During accumulation of data, display is provided prior to and after analysis of each event.
ADD/SUBTRACT — establishes whether data will be added to or subtracted from that previously stored in the System Memory.

TEST — provides a programmed add-one input to each channel of the memory interrogated during display operation. Add-one or subtract-one can be selected by means of the ADD/SUBTRACT Control.

PHA/MCS — PHA establishes the logical programming for pulse height analysis, MCS; the logical programming for multichannel scaling.

SINGLE/MULTI — SINGLE determines that, during display operations only one or two selected memory groups from the X-Axis channels will be interrogated; during readout, only one selected memory group will be interrogated; MULTI determines that all channels will be interrogated.

INTERNAL/EXTERNAL — INTERNAL enables manual selection of certain control functions, i.e., Analyze, Stop, Readout, Add/Subtract, and Data Reduce. EXTERNAL enables these functions to be externally programmed.

DIGITAL BLANK — increases the rate of display to provide optimum viewing of data digitally selected for area of interest intensification.

DATA REDUCE — initiates operation, as determined by the INTEGRATE/DIFFERENTIATE // TRANSFER/STRIP Control.

INTEGRATE/DIFFERENTIATE//TRANSFER/STRIP Control — INTEGRATE/DIFFERENTIATE enables selection of contiguous channels for either integration or differentiation of data stored within those channels. TRANSFER/STRIP enables data from a selected memory group to be multiplied by a constant, and then added to or subtracted from data in another memory group.

A multiplication constant indicator is provided by the first channel of each transfer group. The multiplication constants provided are 10, 1, 0.1, 0.01, and 0.001. Transfer also enables data read in from external sources to be added to, or subtracted from a selected memory group.

Memory Group Selection

Group size can be selected in binary steps (2^n) from 64 to the maximum number of channels provided by the System Memory.

Channel Selection

One channel or a selected band of channels in each memory group can be selected for digital intensification during display operation.

Arithmetic Decoding Resolution

In linear display, nine-bit decoding of digital arithmetic data provides resolution of up to 1 part in 512. Display ranges are provided in binary steps (2^n) from 256 to the maximum capacity of the System Memory.

Display Overlap

Stored data in two selected memory groups can be overlapped and normalized.

Automatic Programming

A control is provided for selection of three different program sequences:

1. Accumulate data for a preset time and then stop.
2. Accumulate data for a preset time and then automatically read out stored data nondestructively.
3. Accumulate data for a preset time, read out stored data destructively, and then repeat the same sequence until stopped by the operator.

Preset Live Times

For data accumulation in pulse height analysis, preset live time units are provided in binary steps (2^n) from 1 to 16 for an 18-bit memory word length or from 1 to 64 for a 20-bit memory word length. Live time units can be from .1 to 10K minutes, or some other externally determined increment, as established by the Data Acquisition Section of the system. A setting of infinity on the Preset Live Time Control enables data acquisition to continue indefinitely until stopped by the operator.

Logarithmic Display

Logarithmic display, either with or without a decade calibration pattern, can be selected. Decade calibration levels up to the most significant digit of maximum memory capacity are generated internally to facilitate display calibration.

"Scan Master" Display Expansion

A selected portion of the display can be expanded horizontally by a factor of ten to twenty as determined by a factory set level. Only the selected portion of the display is scanned to provide optimum viewing.

Memory Erasure Interlock

The memory, or selected portions thereof, can be cleared to zero only when two electrically interlocked pushbuttons are pressed simultaneously. This prevents inadvertent loss of data when one pushbutton is contacted accidentally. An indicator which physically separates the pushbuttons is illuminated when both are pressed.

Display Oscillators

Two display oscillators are included: (1) an address advance oscillator and (2) a shifting oscillator for logarithmic and linear display of arithmetic data.

Decade Pattern Generator

A decade pattern generator is included for calibration purposes in logarithmic display.

Digital to Analog Converters

Circuitry for decoding digital address and arithmetic data to an analog equivalent for display and readout operation is included. Decoded analog voltages are supplied via driver amplifiers.

"Scan Master" Generator

The circuitry associated with the generation of "Scan Master" horizontal display expansion is included.

Memory Group & Channel Select Gates

Logical gating circuitry for memory groups and channel region of interest selection is included.

Display Selection Control Gates

Logical gating circuitry for selection of various display operations is included.

Display Overlap Generator

The circuitry associated with the generation of display overlap for two selected memory groups is included.

Punched Tape Short/Long Leader

The circuitry associated with the generation of either a long or short leader for punched paper tape readout is included.

Lamp Drivers

Lamp driver circuitry for the illuminated indicators is included.

Address Identification Generator

Circuitry associated with the identification of memory group and channel region of interest for digital readout is included.

Printed in U.S.A. 3/67